# Markdown Renderer Test 3: Stress Test and Edge Cases

This file targets the things renderers commonly get wrong. Some items are intentionally tricky. Read the "Expected" note under each section.

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## 1. Code Blocks That Contain Other Syntax

Code containing math delimiters must NOT be rendered as math:

```
price = "$5.00"
cost = "$10 and $20"
formula = r"$$ x^2 + y^2 = z^2 $$"
template = f"Total: ${total}"
```

Code containing a mermaid fence must display as plain text:

```
![Mermaid diagram 1](diagrams/mermaid-bf39c59a9e3f4cb9.pdf)
```

Code containing markdown links and headings:

```
# Not a real heading
[Not a link](https://example.com)
**Not bold**
```

Inline code with special characters: `` `backticks` inside ``, `<div>`, `$x^2$`, `\frac{a}{b}`.

Long line (should scroll or wrap, not break the layout):

```
curl -X POST "https://api.example.com/v1/resources?filter=active&sort=created_at&order=desc&limit=100&offset=0" -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" -d '{"name":"a very long payload","description":"this line is intentionally long to test horizontal overflow handling in the renderer"}'
```

Unknown language and odd info strings:

```
this language does not exist; it should fall back to plain text
```

```
js {1,3} title="example.js" const a = 1; const b = 2; const c = 3;
```

Unicode and emoji inside code:

```
# 日本語のコメント, émojis 🚀, Ελληνικά
greeting = "こんにちは 🌍"
```

Tabs vs spaces:

```
func main() {
    if true {
        println("tab indented")
    }
}
```

**Expected:** no math or diagrams are rendered inside any of these blocks; everything appears as literal text.

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## 2. Dollar Signs: Money vs Math

Prices must stay as text: it costs $5 and the other costs $10, so the total is $15.

Mixed case: the item is $20, and its area is $A = \pi r^2$ square meters.

Escaped dollar: $100 should show a literal dollar sign, not start math.

Adjacent math: $a$, $b$, and $c$ are variables; $a + b = c$.

Spaces inside delimiters (many renderers refuse to treat these as math): $ x $.

**Expected:** "$5 and the other costs $10" stays plain text. `$A = \pi r^2$` renders as math.

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## 3. Math in Awkward Places

**In a table**

| Name | Formula | Value |
|---|---|---|
| Pythagoras | $a^2 + b^2 = c^2$ | classic |
| Golden ratio | $\varphi = \frac{1+\sqrt{5}}{2}$ | $\approx 1.618$ |
| Pipe in math | $\lvert x \rvert$ | absolute value |

**In a list**

1. First: $f(x) = x^2$
2. Second: $f'(x) = 2x$
   - Nested: $f''(x) = 2$
   - Nested block math:

$$\int_0^1 x^2 \, dx = \frac{1}{3} \tag{1}$$

3. Third: $\sum_{k=0}^{n} k = \frac{n(n+1)}{2}$

**In a heading**

### Heading with math: $E = mc^2$

**In a blockquote**

> The Gaussian integral:
>
> $$\int_{-\infty}^{\infty} e^{-x^2} \, dx = \sqrt{\pi} \tag{2}$$

**In bold and italics**
**Bold with math $x^2$** and *italic with math $y^2$*.

**Special LaTeX commands**

$$\begin{aligned} \mathcal{L}(\theta) &= \prod_{i=1}^{N} p(x_i \mid \theta) \\ \log \mathcal{L}(\theta) &= \sum_{i=1}^{N} \log p(x_i \mid \theta) \\ \hat{\theta}_{\text{MLE}} &= \arg\max_{\theta} \log \mathcal{L}(\theta) \end{aligned} \tag{3}$$

$$\mathbf{F} = m\mathbf{a}, \qquad \hbar\omega = E, \qquad \binom{n}{k} = \frac{n!}{k!(n-k)!}, \qquad \overbrace{1+2+\cdots+n}^{n \text{ terms}} \tag{4}$$

$$\left\langle \psi \,\middle|\, \hat{H} \,\middle|\, \psi \right\rangle = E \tag{5}$$

**Expected:** math renders inside tables, list items, headings, and quotes. Multi-line `align` aligns on `=`.

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## 4. Links: Tricky Cases

- Parentheses in URL: Wiki
- Angle-bracket URL with spaces: Spaces
- Fragment and query: Anchor
- Link text with brackets: [bracketed]
- Link wrapping inline code and emphasis: **bold `code` link**
- Same URL in many forms: https://example.com, https://example.com, https://example.com
- Trailing punctuation should not be swallowed: see https://example.com/page, or https://example.com/other.
- Footnote-style reference: ref link, and an implicit reference: ref-1
- Heading anchor links: Section 1, Section 5
- Dangerous scheme (should NOT be clickable or should be sanitized): click me
- Non-http schemes: mail, phone
- Link in a code span should NOT be linked: `[text](https://example.com)`

**Expected:** all valid links clickable, the `javascript:` link is neutralized, and the code-span link stays plain.

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## 5. Mermaid Edge Cases

### Special characters and quoted labels

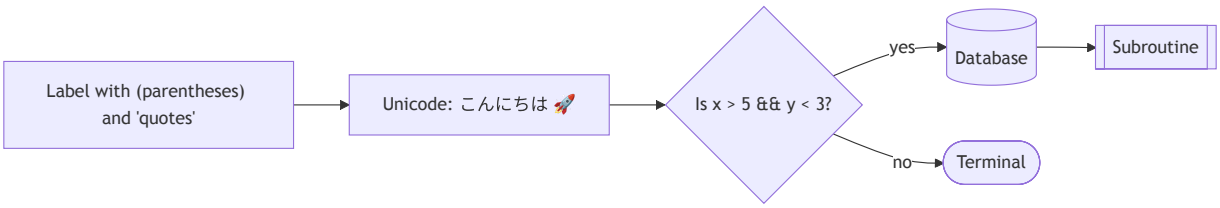



*Figure 1: Mermaid diagram 2*

### Subgraphs and styling

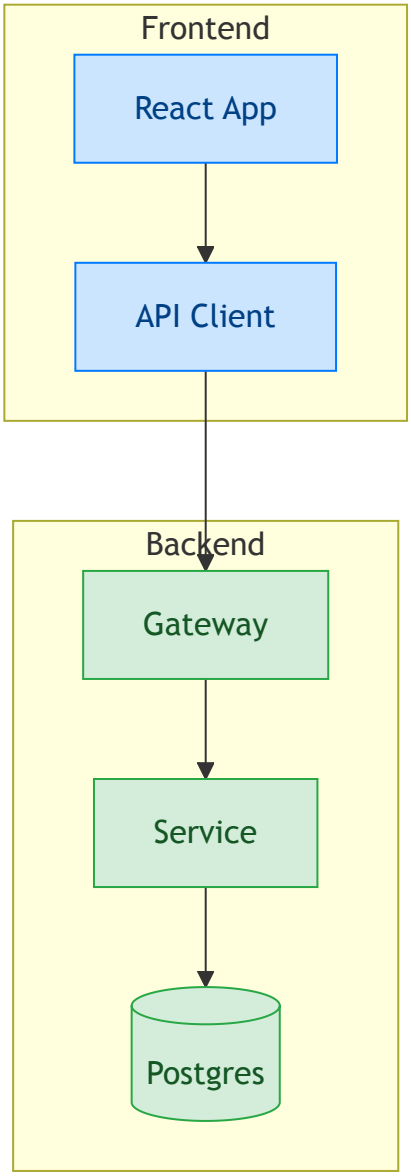



*Figure 2: Mermaid diagram 3*

### Larger sequence diagram with loops and notes

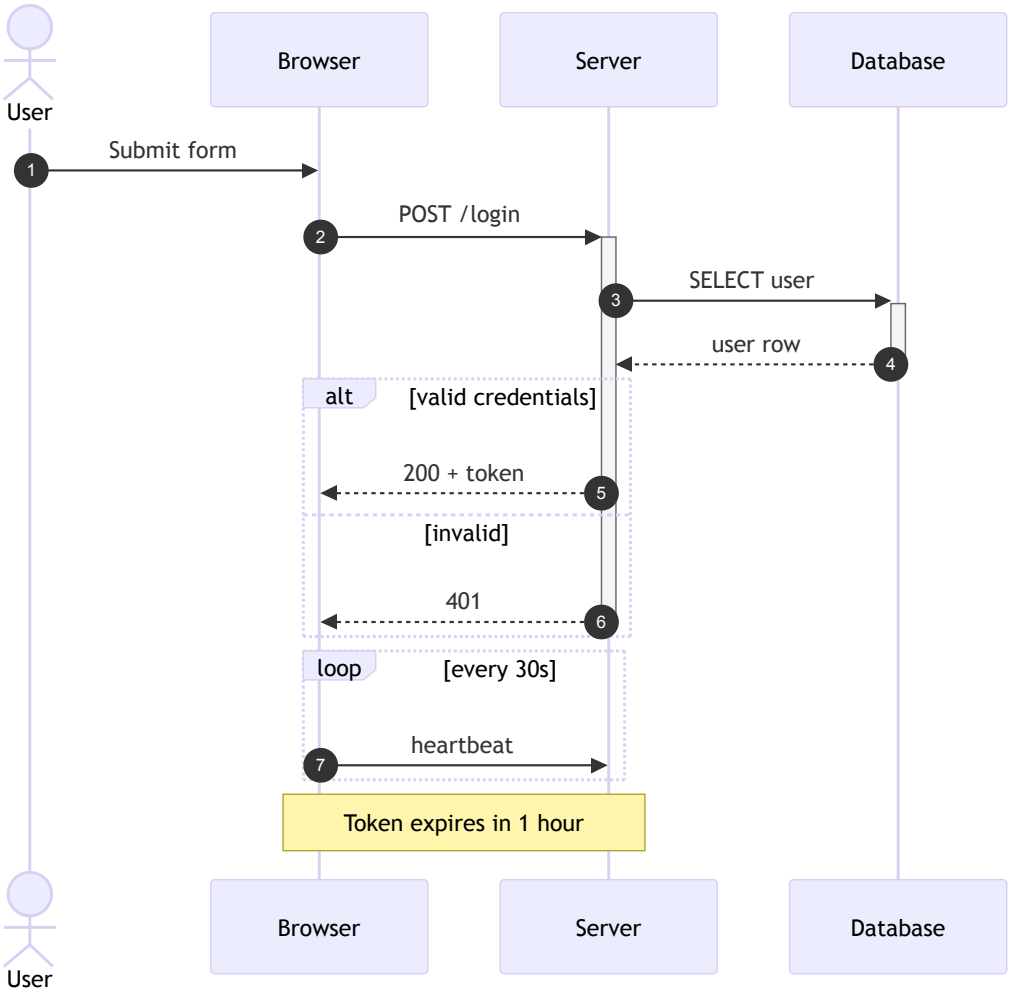



*Figure 3: Mermaid diagram 4*

### Git graph and mindmap

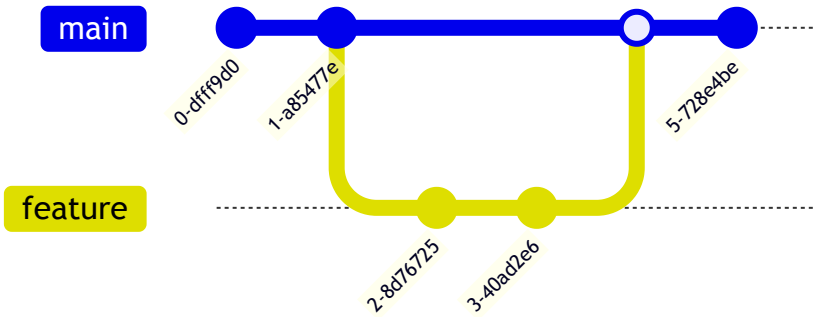



*Figure 4: Mermaid diagram 5*

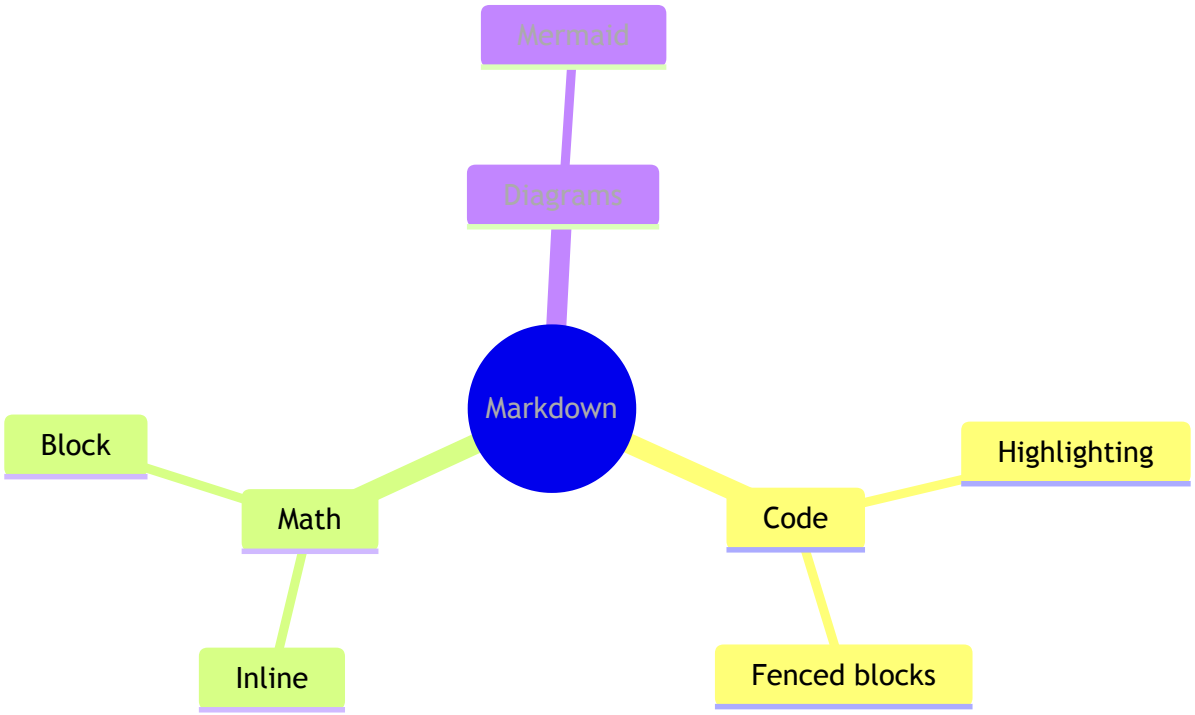



*Figure 5: Mermaid diagram 6*

**Intentionally broken diagram**
The next block has a syntax error. A good renderer shows a readable error message or falls back to the source, and the rest of the page still renders correctly.

```
flowchart TD
    A --> --> B
    this is not valid mermaid
```

**Expected:** all valid diagrams render, with colors and subgraphs intact. The broken diagram fails gracefully without breaking anything below it.

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## 6. Content After the Broken Diagram

If you can read this, the renderer survived the error. The final formula should still render:

$$e^{i\theta} = \cos\theta + i\sin\theta \tag{6}$$

And this final code block should still be highlighted:

```
print("Still alive after the broken diagram!")
```

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## 7. Misc Edge Cases

Escaped characters: *not italic*, _not italic_, `not code`, # not a heading,

$$notalink \tag{7}$$

(url).

Underscores in words: snake_case_variable_name should not become italic.

Asterisks in math: $a*b$ and $a*b$.

Underscore-heavy math (the classic emphasis conflict): $x_1 + x_2 + ... + x_n = \sum_i x_i$.

HTML passthrough: Ctrl + C, H2O, x2, highlighted.

Collapsible section (click to expand) Hidden content with `code`, a link, and math $x^2$.

```
print("code inside details")
```

Horizontal rules in different forms:

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---

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Hard line break with two trailing spaces:
second line.

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## Final Expected Results

- ☐ Code blocks never trigger math or Mermaid rendering
- ☐ Prices with `$` are not mangled into math
- ☐ Math renders in tables, lists, headings, blockquotes, bold, and italic
- ☐ Underscores in math ($x_1 + x_2$) are not eaten by italics
- ☐ Tricky links work; `javascript:` is neutralized
- ☐ Mermaid special characters, subgraphs, styling, gitGraph, and mindmap render
- ☐ The broken Mermaid block fails gracefully and does not stop later content
- ☐ `<details>` collapses and its contents render when opened